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(ସରକାରୀ ଯାତ୍ରିକ ମହାବିଦ୍ୟାଳୟ)

Research Sensitization & Innovative Project Development

Smart Agriculture: A Cognitive Digital Twin for Crop Yield and Failure Detection

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DOMAIN & DESCRIPTION

DOMAIN

- Agricultural Technology (AgriTech)
- Artificial Intelligence
- Cognitive Digital Twin (CDT)

Core Focus : Building a continuous, data-driven feedback loop for proactive crop failure detection.

PROBLEM DESCRIPTION

Currently, agricultural yield forecasting relies heavily on retrospective government statistics that lack real-time predictive capabilities. When climate-induced anomalies—such as severe droughts, extreme thermal stress, or flooding—occur, traditional regression models fail to accurately classify the probability of a crop failure in time for intervention. Furthermore, uninterpretable deep learning models provide predictions without actionable explanations, reducing trust among stakeholders. There is a critical need for an intelligent system that can continuously ingest environmental telemetry, model biophysical failure triggers, account for weather uncertainty, and provide explainable AI insights. A Cognitive Digital Twin (CDT) solves this by replicating the physical agricultural state digitally, allowing for proactive decision-making and counterfactual "What-If" climate simulations.

LITERATURE REVIEWS

PAPER -1 : Climate Prediction Using RNN LSTM to Estimate Agricultural Products Based on Koppen Classification (Andini & Utomo, 2021)

1. **Brief Description:** This study focuses on predicting climate variables to optimize agricultural product estimation and mitigate the risk of crop failure due to incompatible climate conditions.
2. **Dataset Used:** A climate time-series dataset structured around the Köppen climate classification system.
3. **Proposed Methodology:** The researchers implemented a Recurrent Neural Network (RNN) architecture utilizing Long Short-Term Memory (LSTM) cells (with 48 input neurons), optimized via the Adam optimizer over 200 epochs.
4. **Key Outcomes:** The methodology successfully achieved high predictive accuracy, yielding a Mean Absolute Percentage Error (MAPE) of just 3.29% for 1-month-ahead climate forecasting.
5. **Research Gaps:** The model's temporal depth is limited to forecasting just one month ahead, rather than projecting a full 12-week agricultural season. Additionally, it omits critical sub-surface environmental features like root zone soil wetness (GWETROOT), which are essential for robust biophysical stress modeling.

PAPER -2 : A Time-Series Hybrid Multi-Model Machine Learning Framework for Staple Crops Yield Prediction (Arya et al., 2026)

1. **Brief Description:** This research aims to improve the accuracy of staple crop yield forecasting by addressing the complex, nonlinear dynamics inherent in agricultural data.
2. **Dataset Used:** International macro-level agricultural yield datasets and historical time-series production records.
3. **Proposed Methodology:** The authors designed a Time-Series Hybrid Multi-Model Machine Learning Framework, combining traditional linear time-series forecasting (ETS) with non-linear deep learning (Artificial Neural Networks and LSTM).
4. **Key Outcomes:** The hybrid approach demonstrated superior accuracy for staple crops, specifically achieving an R^2 of 98.4% for Rice yield predictions.
5. **Research Gaps:** The framework operates as an uninterpretable "black box," lacking Model Interpretability to explain why a crop might fail. Furthermore, its reliance on international macro-data makes it poorly suited for district-stratified, localized interventions for smallholder farmers.

PAPER -3 : Precision Agriculture Revolution: Integrating Digital Twins and Advanced Crop Recommendation for Optimal Yield

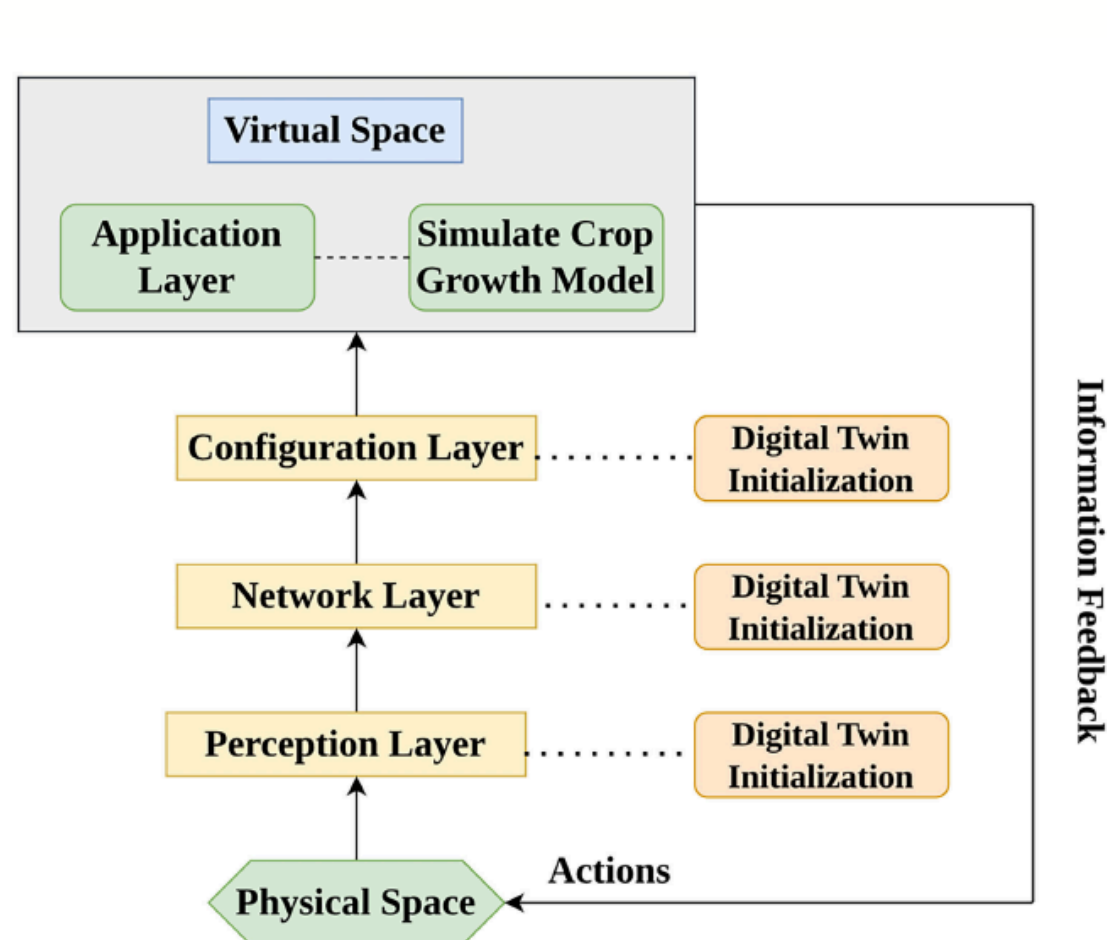
1. **Brief Description:** This paper explores the integration of Cyber-Physical Systems (CPS) to construct an agricultural digital twin capable of advanced, dynamic crop recommendation.
2. **Dataset Used:** Operational farm-level data collected via localized Internet of Things (IoT) sensor networks.
3. **Proposed Methodology:** The researchers developed a digital twin methodology that digitally mirrors the physical farm environment, enabling the execution of dynamic "What-if" simulations for agricultural planning.
4. **Key Outcomes:** The implementation of the digital twin led to highly impactful optimizations, demonstrating a 10–25% increase in crop yield and a 20–30% reduction in water usage.
5. **Research Gaps:** The heavy reliance on a sensor-dense "single-farm silo" creates a massive infrastructure cost and data interoperability bottleneck, rendering it financially unviable for smallholders. It also lacks an automated, explainable mechanism for translating complex AI logic to the farmer.

PAPER -4 : Crop Yield Time-Series Data Prediction Based on Multiple Hybrid Machine Learning Models (Yan et al., 2025)

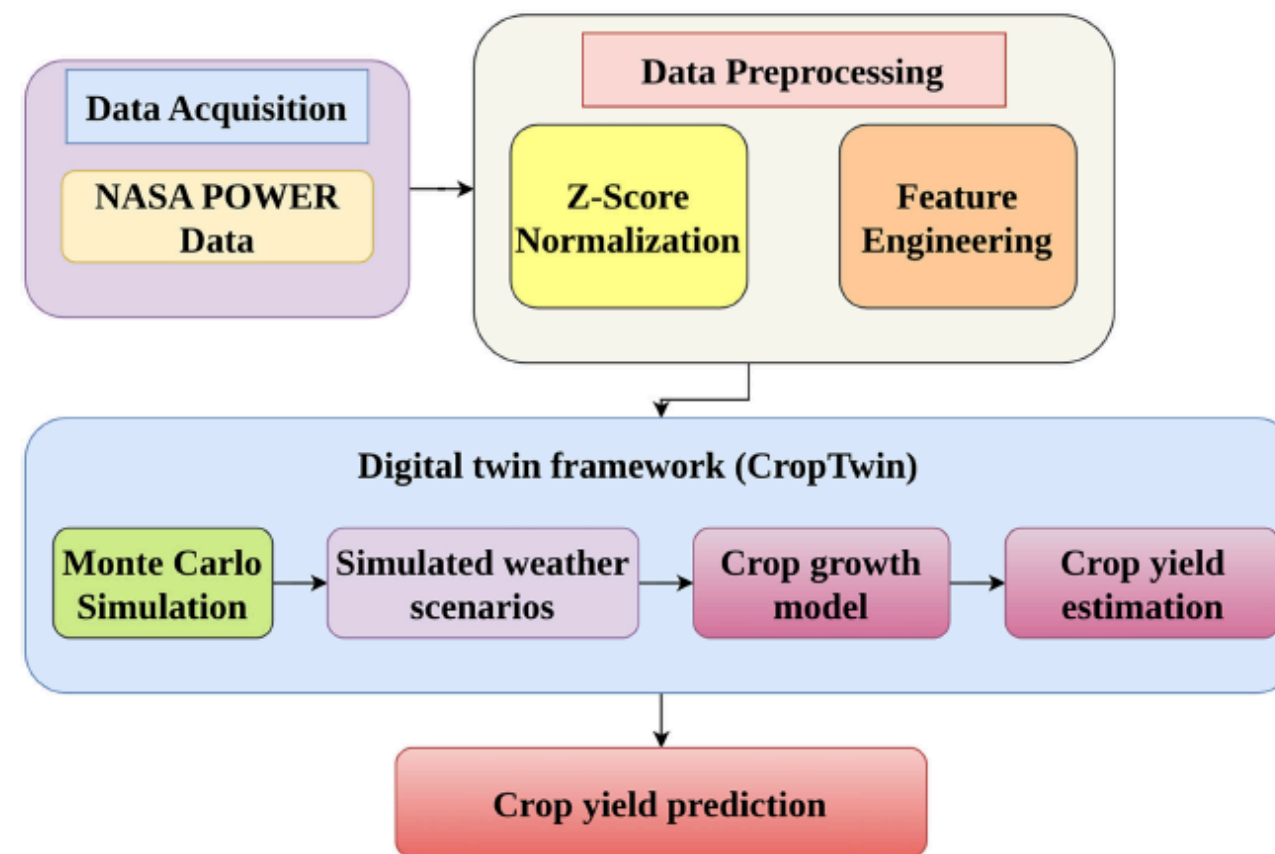
1. **Brief Description:** This study focuses on enhancing the accuracy of continuous crop yield predictions by analyzing complex, non-linear relationships in agricultural time-series data.
2. **Dataset Used:** Historical agricultural yield metrics combined with localized, real-time pesticide usage records.
3. **Proposed Methodology:** The researchers constructed a Weighted Ensemble model that combines the predictive strengths of Random Forest (RF) and XGBoost algorithms.
4. **Key Outcomes:** The hybrid ensemble approach significantly outperformed standalone models, successfully reducing the Mean Absolute Error (MAE) from 0.0177 (standalone RF) to 0.0042.
5. **Research Gaps:** The model strictly predicts continuous yield volume (t/ha) rather than classifying the binary "Failure Anomalies" required for disaster relief triggering. Additionally, its heavy reliance on real-time pesticide data creates a severe operational bottleneck in developing regions where such data is often delayed or unavailable.

PAPER -5 : Digital Twin-Based Uncertain Weather Condition Monitoring for Enhanced Crop Yield Prediction (Kenneth, 2026)

1. **Brief Description:** Kenneth (2026) introduces the "CropTwin" framework, designed to monitor and mathematically model uncertain weather conditions to enhance the prediction of crop yields without relying on local hardware.
2. **Dataset Used:** Continuous satellite meteorological telemetry sourced directly from the NASA POWER API.
3. **Proposed Methodology:** The framework integrates digital twin technology with Monte Carlo simulations to computationally model weather uncertainty and evaluate counterfactual environmental scenarios.
4. **Key Outcomes:** The simulated uncertainty approach proved highly viable, generating a real-time yield prediction of 5.90 t/ha compared to an uncertain weather projection of 5.62 t/ha using only satellite API data.
5. **Research Gaps:** The framework exclusively models weather uncertainty while completely ignoring critical biotic stress factors (like pest infestations driven by humidity). It also operates purely as a predictive monitor, lacking a Decision Support System (DSS) to issue actionable intervention advisories back to the physical layer.



The digital twin framework and functions of each layer.



Schematic representation of the proposed CropTwin digital twin-based crop yield prediction framework.

Key Limitations in Existing Literature (Research Gaps)

Prediction of Continuous Yield vs. Binary Disaster Risk

- Existing hybrid models (e.g., Yan et al., 2025) focus entirely on predicting absolute crop volume (t/ha).
- They fail to classify whether a drop in yield constitutes a full-scale agricultural "failure anomaly," which is the actual metric required for immediate disaster intervention.

The Model Interpretability Gap

- Advanced deep learning models (e.g., Arya et al., 2026; Andini & Utomo, 2021) achieve high accuracy but operate as uninterpretable black boxes.
- They do not map mathematical predictions (like LSTM attention weights) to real-world biophysical triggers (such as severe drought or thermal sterility), reducing stakeholder trust.

High Infrastructure Cost and Scalability Bottlenecks

- Current Cyber-Physical Systems (CPS) and agricultural Digital Twins often rely heavily on expensive, localized IoT sensor networks.
- This "single-farm silo" approach poses massive data interoperability challenges and is financially unviable for smallholder farmers in regions like Odisha.

Omission of Actionable Feedback Loops

- While systems like Kenneth's (2026) CropTwin successfully model weather uncertainty, they operate purely as passive predictive monitors.
- They lack an integrated, intelligent Decision Support System (DSS) that automatically translates complex predictions into actionable agronomic advisories for the end-user.

Incomplete Environmental Stress Profiling

- Current models often rely on high-friction, hard-to-acquire data like real-time pesticide usage, while omitting critical sub-surface metrics like Root Zone Soil Wetness (GWETROOT).
- This causes predictions to fail when chemical application data is delayed or when complex biotic stress triggers are ignored.

Our Innovative Approach (The Cognitive Digital Twin Framework)

Dual-Track ML Engine for Binary Failure Classification

Implements a dual-track engine (Tabular ML alongside a PyTorch 3D Tensor LSTM-Attention model).

- We utilize a probabilistic Q1 thresholding logic on the data to explicitly classify, label and predict Failure Anomalies" for disaster relief prioritization.

Macro-to-Micro Data Bridging via NASA POWER API

- To solve the IoT hardware infrastructure bottleneck, our framework completely bypasses the need for expensive local sensors.
- We ingest continuous, district-stratified macro-data directly from the NASA POWER API, computationally transforming satellite weather telemetry into localized field-level risk proxies.

Model Interpretability Mapped to Biophysical Triggers

- We ensure high model interpretability by extracting Temporal Attention Weights from our LSTM network.
- We map these mathematical weights directly to physical agricultural stressors (e.g., automatically flagging "High risk due to week-4 thermal stress").

LangGraph-Powered Expert Advisory Node (DSS)

- Instead of just displaying a raw risk percentage on a screen, we close the feedback loop by embedding an AI-driven Chatbot directly into the Application Layer.
- This Orchestrator Agent automatically translates complex tensor outputs into plain-language, actionable interventions and advisories tailored for the farmer.

Interactive "What-If" Climate Simulations

- We elevate the system from a passive "Digital Shadow" to a true Cognitive Digital Twin by enabling dynamic counterfactual testing.
- Our dashboard allows analysts to dynamically manipulate climate variables (temperature, humidity) and run Monte Carlo simulations to observe potential future disaster outcomes.

PROBLEM STATEMENT :

To design and develop a Cognitive Digital Twin that integrates 20 years of historical yield data (APY Data by Govt. of Odisha) with continuous NASA POWER weather telemetry to accurately forecast crop yield, classify failure anomalies using an intelligent prediction engine, and provide explainable Decision Support System (DSS) and can run "what-if" simulations.

OBJECTIVES :

- 1.**Engineer an automated data pipeline** : A pipeline that harmonizes two decades of historical crop yield statistics with continuous, satellite-based weather telemetry from the **NASA POWER API**.
- 2.**Formulate a statistical framework**: A framework that accurately defines, quantifies, and labels district-level crop failure anomalies for machine learning tasks.
- 3.**Design and train a dual-purpose predictive engine** : capable of forecasting continuous seasonal crop yields while simultaneously classifying the probability of imminent crop failure.
- 4.**Deploy an intelligent Decision Support System (DSS)** : A natural language query-response interface for easy interaction with the predictive models
- 5.**Construct an interactive Digital Twin dashboard** : A Dashboard that empowers stakeholders to run counterfactual "what-if" climate simulations .

Functional Requirements

High-Level Requirements	Input	Process	Output
1.Ingest Historical Data	Raw UPAg/Govt yield CSVs (1997–2025)	Clean text, map seasons, and standardize yield to Quintals/Acre.	Structured historical database
2.Interpolate Data Gaps	Harmonized database with missing years	Apply time-series interpolation (linear/polynomial) to estimate gaps.	Continuous 20-year yield grid
3.Calculate Thresholds	Continuous 20-year grid	Group by district/season and calculate First Quartile (Q1) thresholds.	Binary Failure_Anomaly labels
4.Fetch Satellite Telemetry	District Coordinates & Dates	Execute API requests to NASA POWER for daily meteorological metrics.	Raw daily weather time-series
5.Merge Telemetry	Yield labels + Daily weather	Aggregate weather into 12-week windows and inner join with yield.	Final ML Dataset (Tabular & 3D)
6.Train Predictive Models	Final ML Dataset	Train Dual-Track router (RF+XGBoost and PyTorch LSTM-Attention).	Serialized model weights (.pt)
7.Predict Failure Risk	Real-time 12-week weather	Run inference and extract LSTM temporal attention weights.	PredictionResult (Yield + Risk)
8.Visualise Summaries	PredictionResult objects	Parse probabilities to render interactive GIS layers and metric cards.	Web Dashboard UI
9.Run What-If Simulations	User-adjusted climate sliders	Apply perturbations to baseline data and run Monte Carlo simulations.	Counterfactual yield & risk metrics
10.Query Expert Chatbot	Natural language questions	LangGraph agent fetches prediction context to prompt an LLM.	Plain-language agronomic advisory
11.Sync Seasonal Data	Automated end-of-season trigger	API call to UPAg to download new real-world ground truth data.	Updated database & retrained model

Non-Functional Requirements

SL NO.	Category	Requirement Summary
1	Reliability	Must guarantee 99.9% uptime and implement auto-retry logic with cached fallbacks in case the NASA POWER API drops connection.
2	Performance	Interactive "What-If" climate simulations must return updated risk predictions to the UI in under 2 seconds.
3	Scalability	The backend architecture must be horizontally scalable to support multi-state expansion beyond the initial Odisha pilot without schema redesigns.
4	Usability	The Expert Advisory Chatbot must translate complex tensor outputs into plain, non-academic language for farmers.
5	Security	All API data transfers must use HTTPS/TLS encryption; role-based access must separate 'Analyst' read/write privileges from 'Farmer' read-only access.
6	Maintainability	The ML engine must be fully decoupled from the UI, allowing data scientists to swap out deep learning models without breaking the dashboard.

Technology Stack

AGENTIC-AI FRAMEWORKS



LangGraph



LangChain

Backend & Orchestration



LangGraph

Frontend & Visualization



--Next-js
UI and
Server



--Typescript
Api
connection

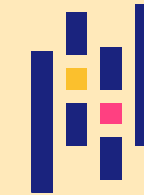


--Tailwind CSS
Design

Data Engineering



Python



Pandas



Numpy



NASA POWER API

ML-DL Framework



PyTorch

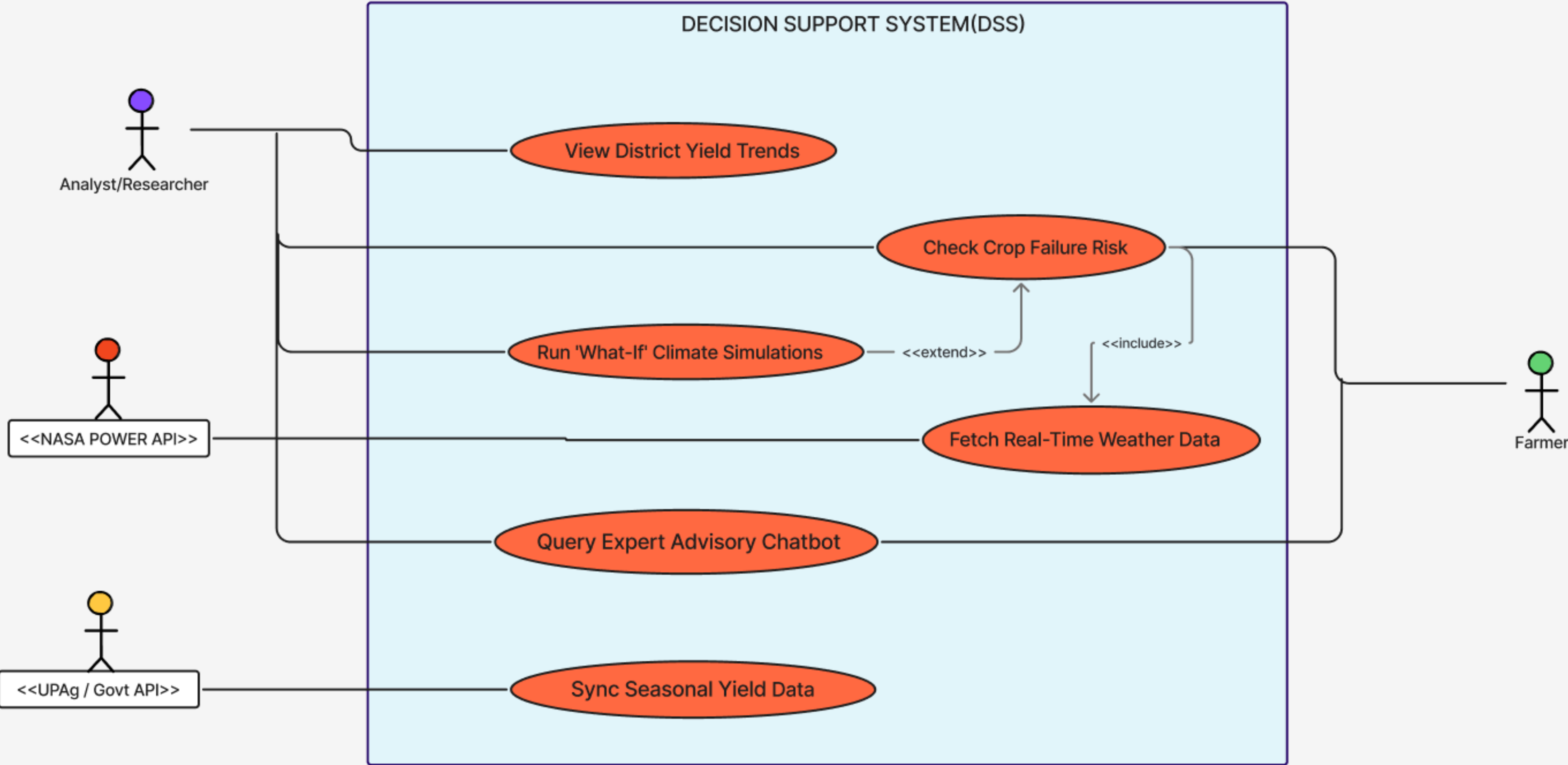


TensorFlow

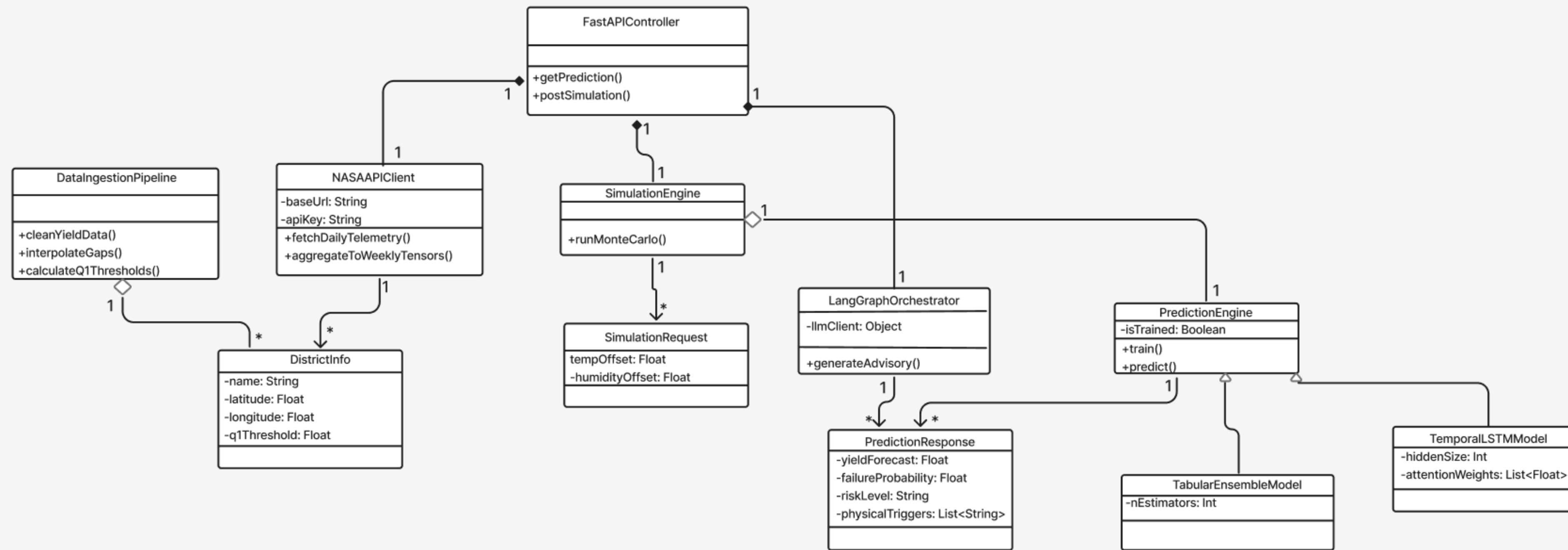


Scikit-Learn

USE-CASE DIAGRAM



CLASS DIAGRAM



Compliances To First Meeting comments

Query -1 : How will a farmer get predictions for their exact farm location without wrong results ?

Implemented Solution : Integrated Leaflet **GIS** map with ray-casting to detect the correct district from coordinates. Added coordinate pinning for field-level selection.

Query -2 : The architecture modeling in UML diagrams was incorrectly represented.

Implemented Solution : All the diagrams now accurately reflect the implemented system.

Query -3 : Progress was shown in normal boxes, not a Gantt chart.

Implemented Solution : Replaced box-based timeline with a proper Gantt chart showing across 6 weeks with phase grouping.

Query -4 : How does the system achieve real-time $t \rightarrow t+dt$ streaming?

Implemented Solution : Built virtual sensor replay with 84-day deque state + WebSocket bridge $\rightarrow$ Live Virtual Twin modal.

Summary of Planning, Analysis & Design Phase

PLANNING

1. **Domain** : AgriTech + AI — Cognitive Digital Twin for Odisha rice yield prediction.
2. **Approach** : ML-DL based (XGBoost + LSTM + Ridge stacking(Ensemble)).
3. **Scope** : 30 districts, 2 seasons (Kharif/Rabi), 20 years (2006-2024).
4. **Data Sources** : Govt yield statistics + NASA POWER satellite telemetry (4 weather vars).
5. **Dual-Target Objective** : Predict continuous yield (Q/Acre) + classify binary failure anomaly.

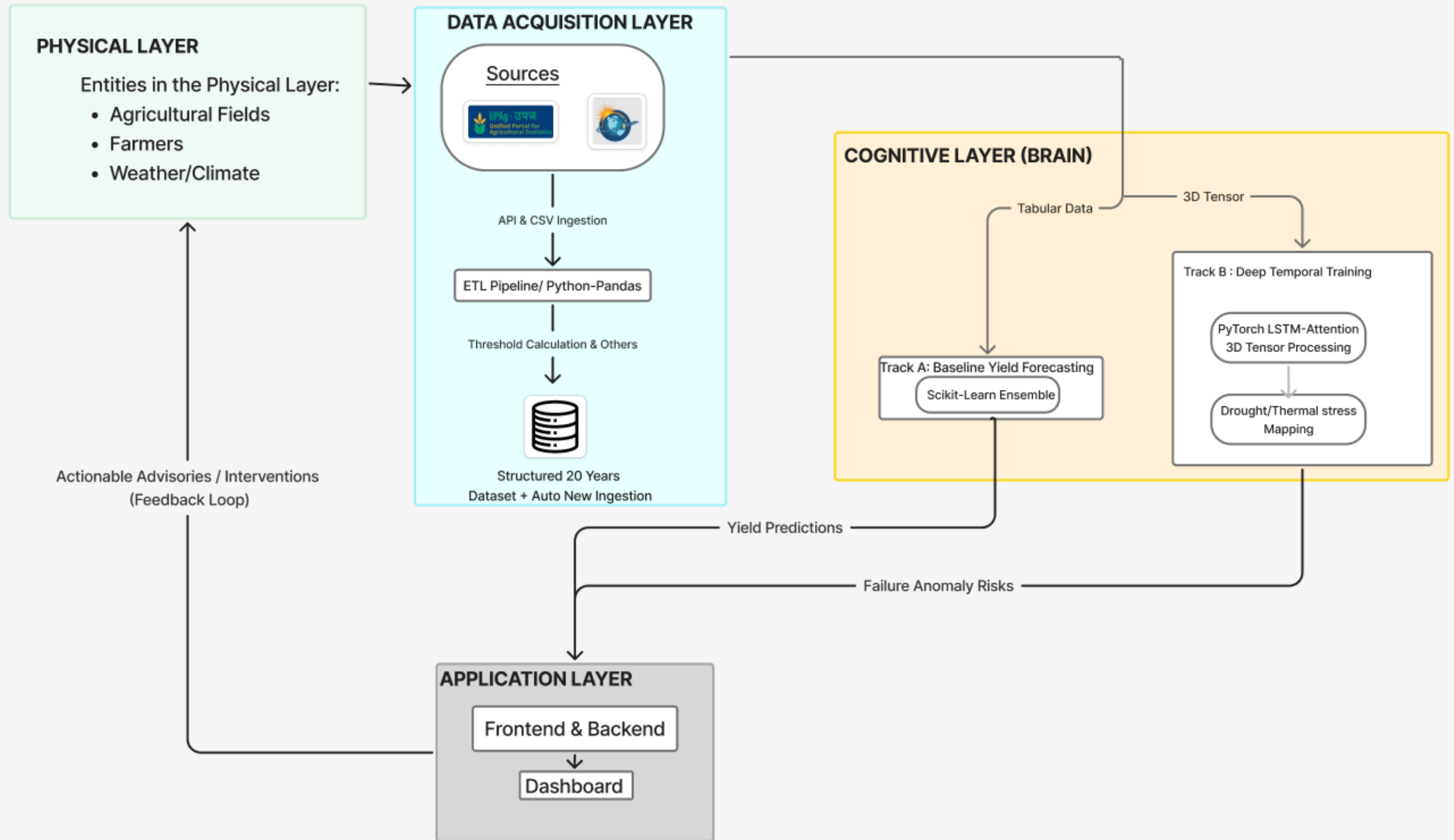
ANALYSIS

1. **Labeling Gap** : Existing research predicts yield volume only; disaster relief needs binary failure classification.
2. **Explainability Gap** : Deep learning is black-box (only output); no mapping from model internals to physical crop stress.
3. **Macro → Micro Gap** : No local sensor infrastructure for smallholders; satellite data must proxy field-level risk.
4. **Real-Time Gap** : Batch systems lack live feedback loop for continuous monitoring.

DESIGN

1. **Data Harmonization** : 84-day window extraction → weekly aggregation → Q_1 threshold labeling → 80/10/10 random split.
2. **Feature Design** : Absolute year offset (year-2006) — preserves temporal generalization.
3. **Model Architecture** : LSTM (84-step attention) + XGBoost (23 features) + Ridge Stacking with bounded [0.2, 0.8] weights.
4. **Routing** : Custom orchestrator — 5 query types route to minimal required model nodes.
5. **Real-Time Layer** : Publisher-Subscriber router + 84-day deque state + WebSocket broadcast.
6. **Tech Stack** : React 19 + Vite 8 | FastAPI + WebSocket | Groq LLM (llama-3.3-70b) | Leaflet GIS | Recharts.

DIGITAL TWIN ARCHITECTURE



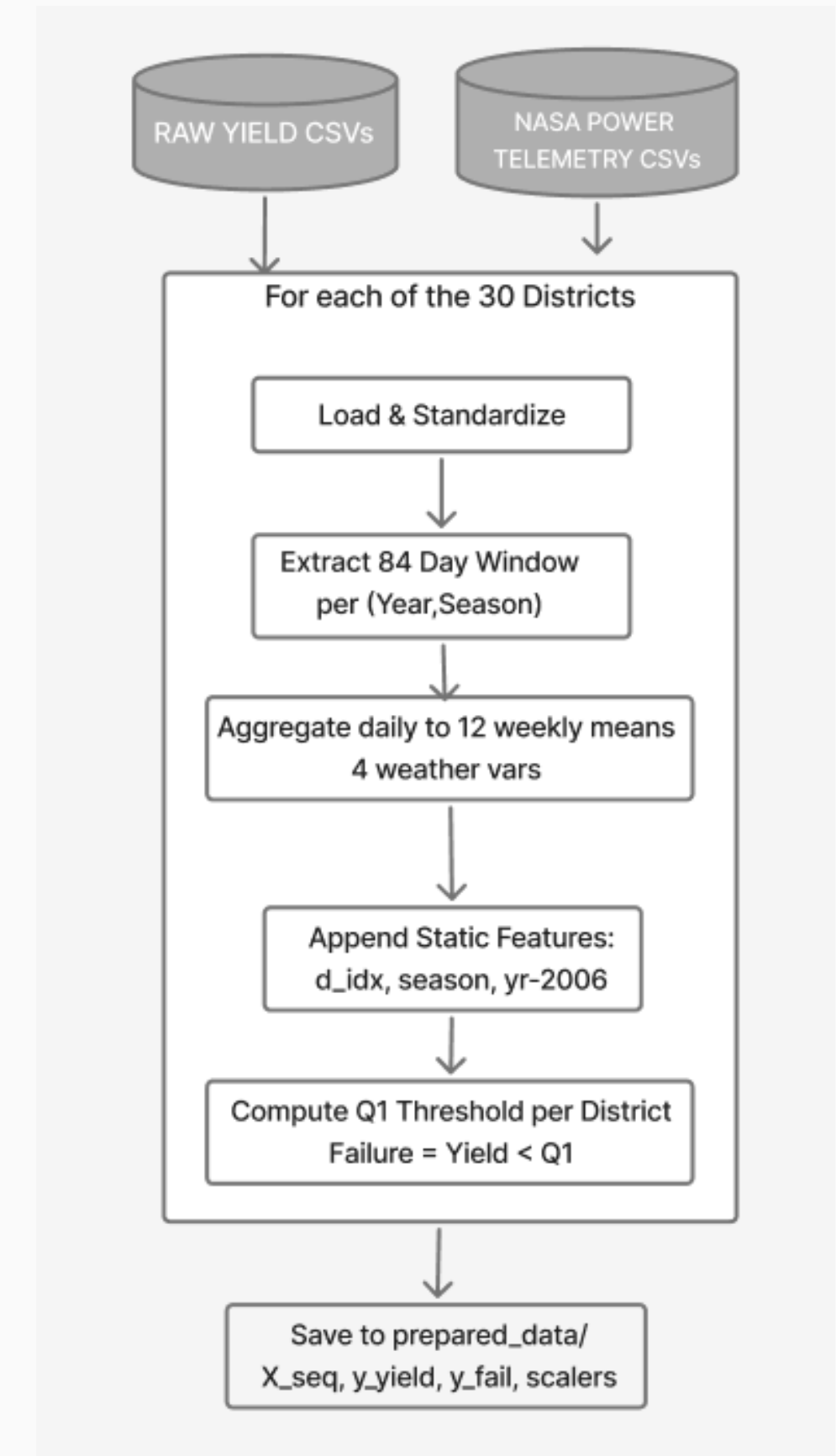
Algorithm 1: Data Ingestion & Harmonization Pipeline

Input: Govt yield CSVs + NASA POWER daily telemetry (4 vars).

Output: Synchronized tensors X_{seq} (1113, 84, 4), y_{yield} , y_{fail} , scalars.

Steps:

1. Extract 84-day window per (district, year, season) — Kharif: Jun 15, Rabi: Nov 1
2. Aggregate 84 daily records → 12 weekly means per variable
3. Append static features: [district_idx, season_onehot, year-2006]
4. Compute Q_1 (25th percentile) threshold per district×season → label failure = yield < Q_1
5. Scale features → save tensors + encoders



Algorithm 2: Digital Twin State Update Algorithm

Input: New yield record(s) for a (district, year, season).

Output: Updated or preserved model state in the deployed twin..

Steps:

- 1. Validate** : New record must have >50% non-interpolated rows; reject if insufficient ground truth.
- 2. Merge** : Combine with existing dataset, fetch missing NASA telemetry for gap years.
- 3. Retrain** : Run full pipeline (LSTM 200 epochs, XGBoost 500 trees, stacking calibration).
- 4. Evaluate** : Compare new model's R^2 and AUC against current deployed model on held-out validation set.
- 5. Decide** : Deploy if new $R^2 \geq \text{prev } R^2 - 0.02$ AND $\text{AUC} \geq \text{prev AUC} - 0.02$; if only failure AUC improved, deploy failure model alone; otherwise fallback to current models.
- 6. Cold-start** : If a new district appears with no history, use nearest-district yield distribution as proxy.

Mathematical Decision Rule

Let M_{old} = currently deployed model with metrics (R^2_{old} , AUC_{old})

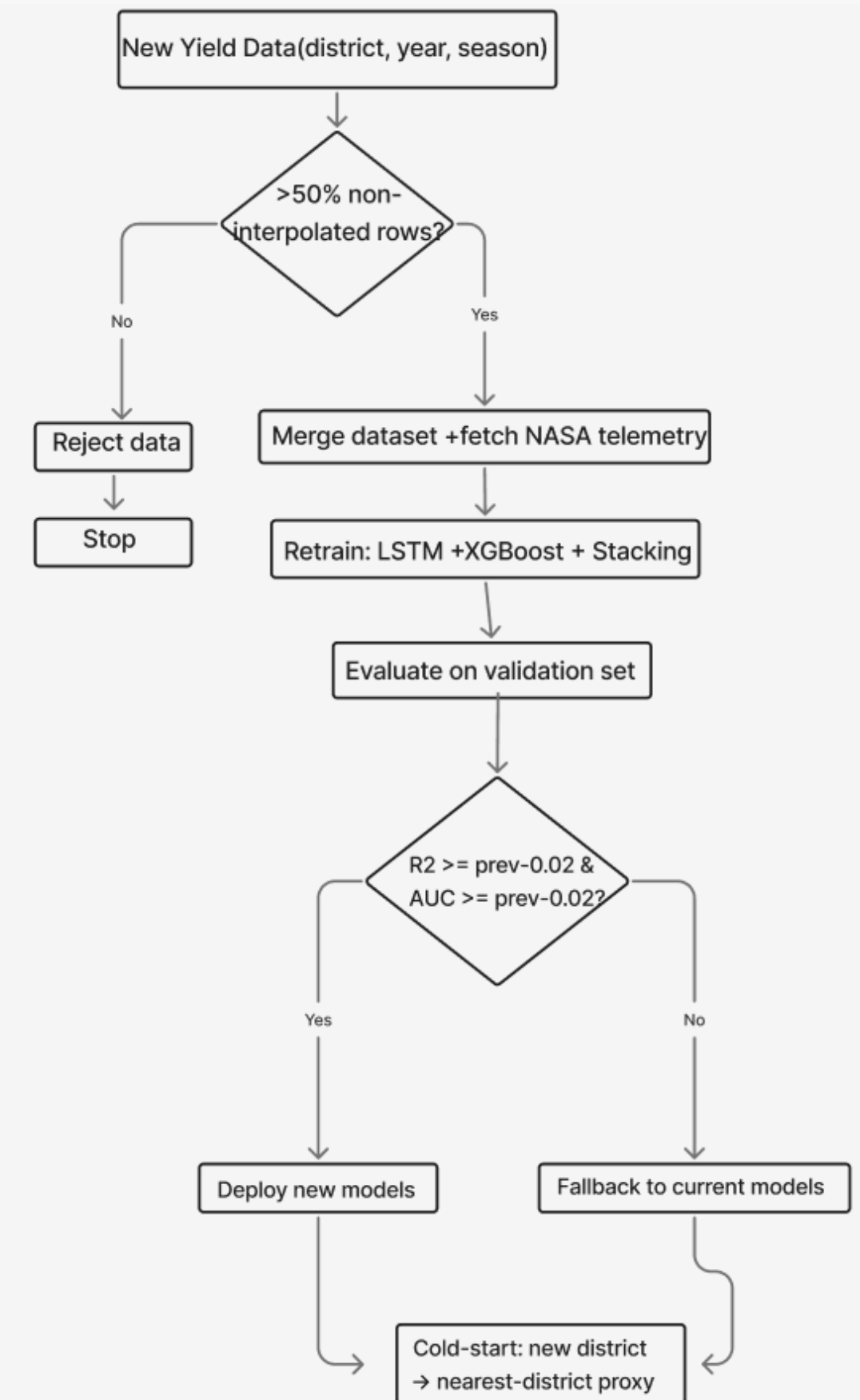
Let M_{new} = retrained model with metrics (R^2_{new} , AUC_{new})

Tolerance $\varepsilon = 0.02$

Deploy M_{new} if: $R^2_{\text{new}} \geq R^2_{\text{old}} - \varepsilon$ AND $\text{AUC}_{\text{new}} \geq \text{AUC}_{\text{old}} - \varepsilon$

Deploy $M_{\text{new_fail}}$ only if: $\text{AUC}_{\text{new}} - \text{AUC}_{\text{old}} \geq \varepsilon$ (failure improved significantly)

Fallback otherwise: keep M_{old} , log regression

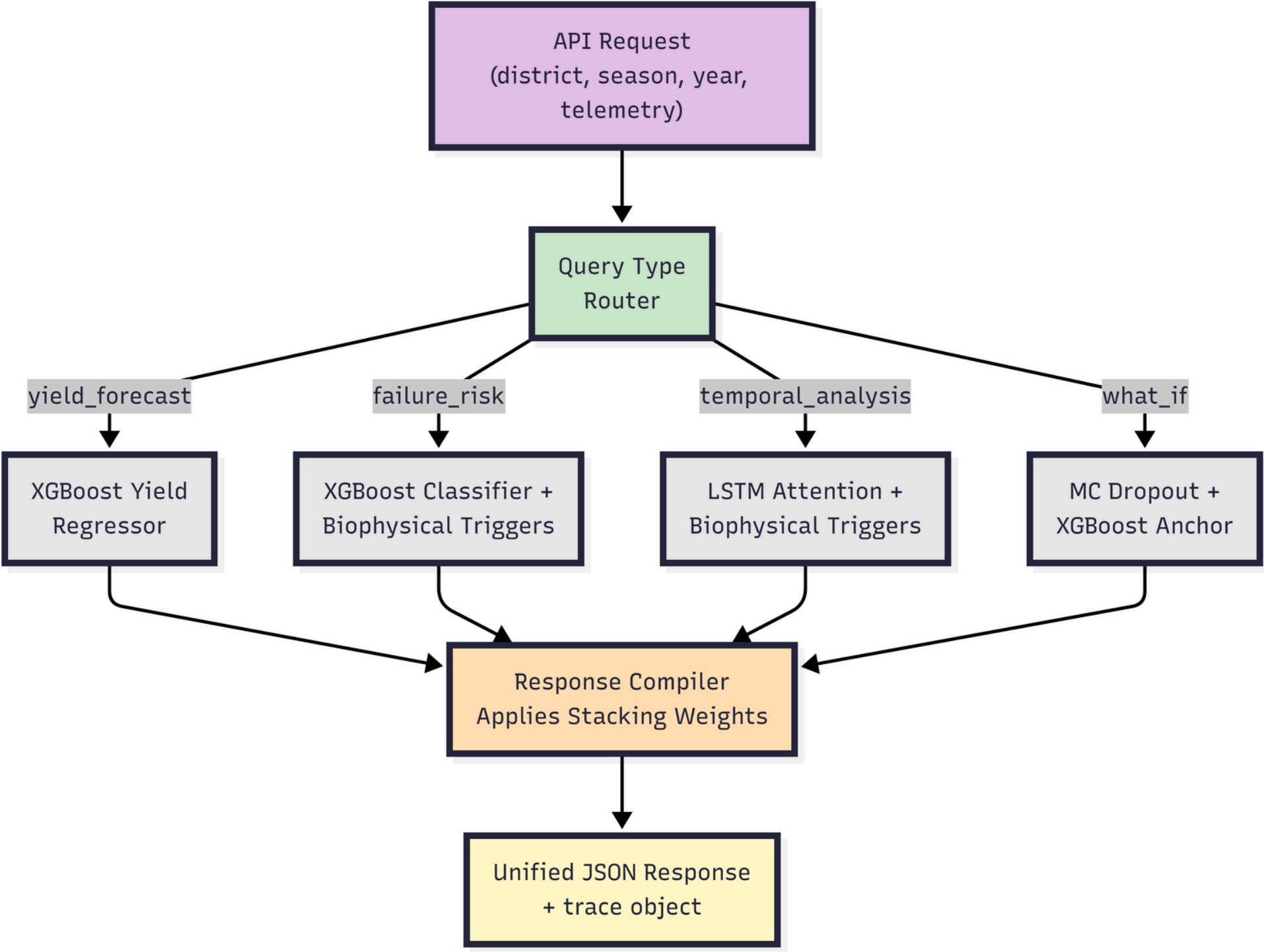


Algorithm 3: Multi-Model Prediction Ensemble

Input: API request with query type .

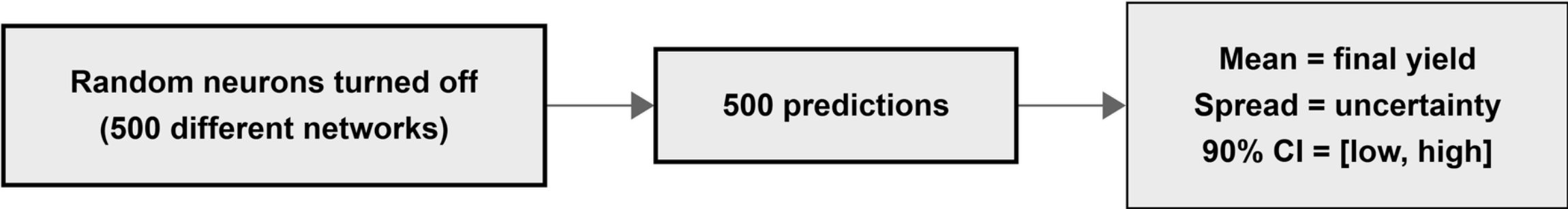
Output: Unified JSON with blended predictions + orchestration trace.

- Steps:**
- 1. Classify intent into one of 5 query types (yield_forecast, failure_risk, temporal_analysis, what_if, full_diagnosis).
 - 2. Route through DAG — activate only the required model nodes.
 - 3. Compute predictions at each activated node
 - 4. Blend using calibrated Ridge meta-learner weights (yield: 0.2L/0.8X, failure: 0.68L/0.32X).



SL NO.	Model	Role
1	LSTM-Attention (84-step, 56K params)	Temporal feature extraction
2	XGBoost Regressor (23 features)	Yield prediction
3	XGBoost Classifier	Failure risk + triggers
4	Ridge Meta-Learner	Blends LSTM + XGB outputs

Algorithm 4: Monte Carlo Dropout for Uncertainty Quantification



Input: Telemetry vector + district/season/year.

Output: Yield prediction with 90% confidence interval + standard deviation.

- Steps:**
- 1. Feed telemetry through LSTM with dropout active.
 - 2. Repeat 500 times → collect 500 yield predictions.
 - 3. Sort them → 90% CI = value at position 25 to value at position 475.
 - 4. Blend each sample with XGBoost (via stacking) for final distribution.

Maths

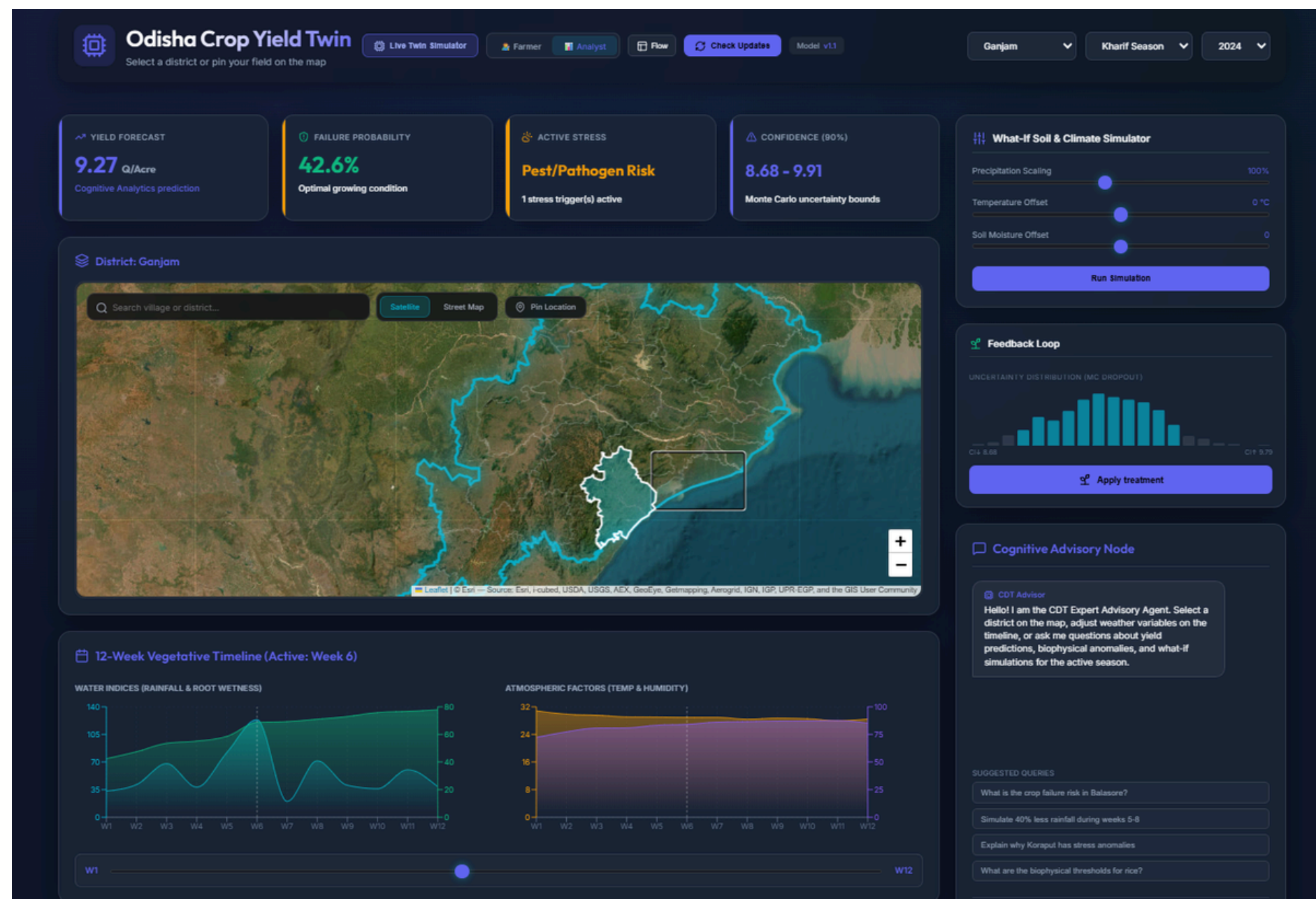
What	How
Each pass	Random 20% neurons disabled, hence different yield
Final prediction	Average of all 500 predictions
Uncertainty	Standard deviation of the 500 predictions
90% Confidence Interval	5th to 95th percentile (sorted)

IMPLEMENTATION AND CODING

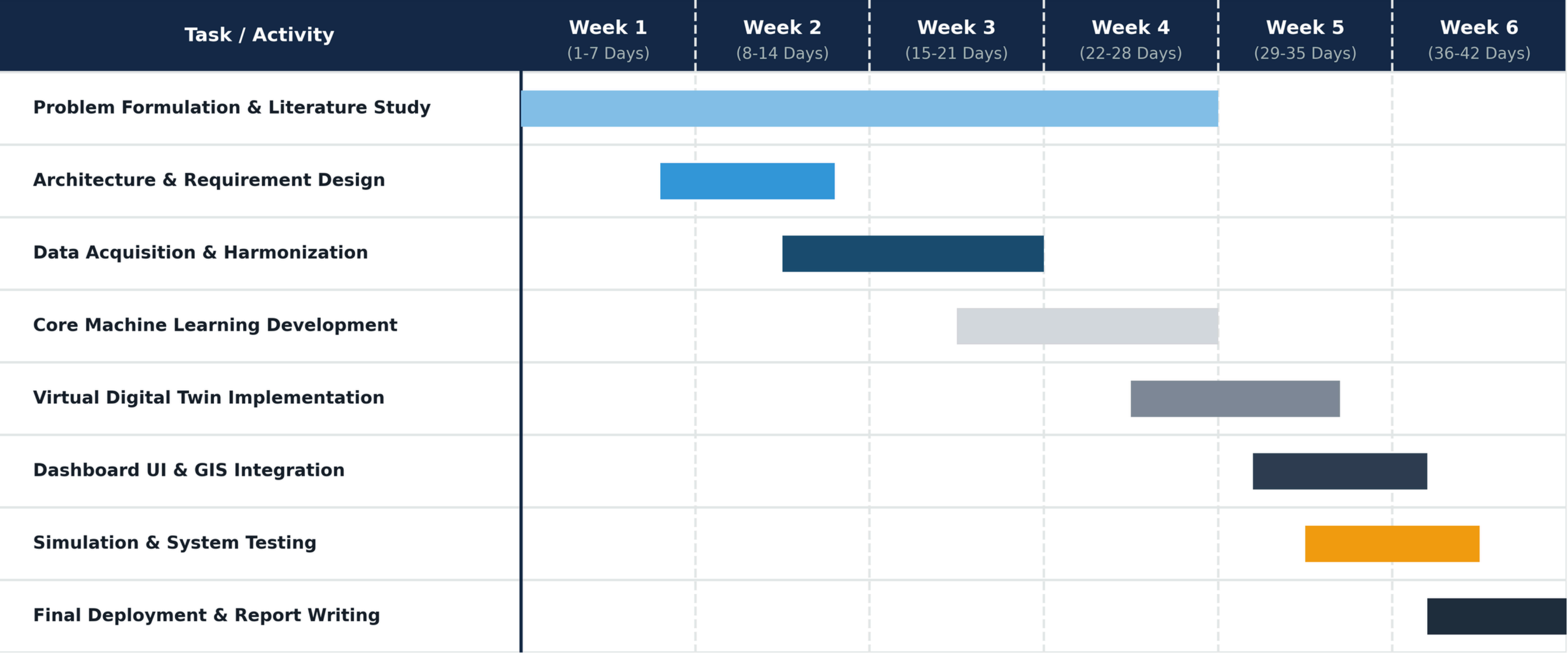
• Codebase Overview

frontend/ React 19 + Vite 8	
▸ App.jsx	Main app, 11 lazy components
▸ components/	GIS map, DSSChat, FeedbackLoop, LiveTwin, dashboards
▸ index.css	Dark glassmorphism theme (~900 lines)
backend/ FastAPI + MQTT + WebSocket	
▸ main.py	13+ REST endpoints + MQTT subscriber + WS manager
▸ llm_client.py	Groq llama-3.3-70b integration
▸ virtual_sensor.py	CSV → MQTT publisher
training-pipeline/ Core ML Pipeline	
▸ prepare_data.py	Data harmonization + feature engineering
▸ train.py	LSTM, XGBoost, Stacking training
▸ predict.py	Inference + MC Dropout + triggers
▸ orchestrator.py	DAG router (5 query types → 5 model nodes)
▸ update_pipeline.py	Full retraining pipeline
▸ models/lstm_final.pth	LSTM-Attention (56K params)
▸ models/xgb_regressor.json	XGBoost yield regressor (23 features)
▸ models/xgb_classifier.json	XGBoost failure classifier
▸ models/meta_yield.pkl	Ridge stacking meta-learner (yield)
▸ models/meta_fail.pkl	Ridge stacking meta-learner (failure)
▸ models/autoencoder.pt	Pretrained masked autoencoder
▸ version.json	Model version + last-run metrics

• Implemented Dashboard



ROADMAP



CONCLUSION

- **The Cognitive Digital Twin framework successfully bridges the critical gap between passive agricultural yield reporting and proactive, predictive disaster management. By continuously integrating 20 years of historical government baseline data with automated NASA POWER satellite telemetry, the system achieves a robust, dual-track machine learning prediction engine.**
- **Crucially, the framework ensures high model interpretability by translating complex deep learning temporal attention weights into transparent biophysical triggers (such as severe drought or thermal stress).**
- **Ultimately, this framework transforms complex data into an interactive Decision Support System (DSS) where stakeholders can run counterfactual "what-if" climate simulations—moving agricultural strategy away from reactive crisis response and toward proactive resilience planning.**

References

- [1] Andini, A., & Utomo, P. (2021). Climate Prediction Using RNN LSTM to Estimate Agricultural Products Based on Koppen Classification. *Journal of Physics: Conference Series*.
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- [5] Sayan Banerjee*, Aniruddha Mukherjee*, Suket Kamboj . (2024). Integrating Digital-Twins and Advanced Crop Recommendation Agricultural Systems.
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